Student Skill Assessment Portal

Project Implementation Architecture & Functional Documentation

1. Relational Database Models & Persistence Schema

The system core utilizes object-relational mapping models configured under Hibernate to manage student records, capabilities, and system testing evaluations securely inside the persistent MySQL instances.

Model Name	Attributes & Properties	System Responsibility / Purpose
User Entity	id (Primary Key) username (Unique) password (Encrypted) email (Unique) role (Student / Admin)	Handles security profiles, manages platform access permissions, and governs distinct user session rights.
Skill Entity	id (Primary Key) skillName (Unique) description (Text Block)	Acts as a master directory for competencies against which structural gaps are checked during student testing.
Assessment Entity	id (Primary Key) student (User Link) roleCategory (String) score (Integer) completedAt (Timestamp)	Persists individual assessment score outcomes, logging performance history records dynamically.

2. Application Architecture Layer Subdivisions

A strictly decoupled three-tier architectural framework governs all transaction pipelines, promoting structural integrity and safe boundaries between operations.

System Layer	Functional Responsibility & Operations
Repository Layer	Extends standard JPA interfaces to safely query, update, and read object records from the persistent tables without handling lower-level SQL scripts explicitly.
Service Layer	Houses transactional boundaries, handles encryption matches, maps entity relational objects, and calculates performance metrics.
Controller Layer	Listens to active browser incoming GET/POST endpoint routes, maps model attributes contextually, and redirects execution flow cleanly to specified view fragments.

3. Structural Dashboard Summary Metrics (Step 4 Rules)

To satisfy explicit evaluation transparency rules, the main profile index page dynamically computes a core metrics matrix detailing the active student status metrics:

Metric Name	Functional Definition	Expected Output Format Example
Current Target Role	Displays the desired career path selection chosen by the student.	Java Backend Developer
Latest Match Score	The structural alignment percentage calculated from the last evaluation run.	46.15%
Matched Skills	Total sum count of profile competencies currently in active dataset alignment.	3 Skills
Missing Skills	Total capability gaps identified against the requirements matrix.	3 Skills (Spring Boot, REST, Hibernate)
Highest Previous Score	Historic personal peak evaluation profile score value attained.	85.00%
Number of Assessments	Total counter tally tracking all historical evaluation submissions.	4 Runs
Top Recommended Skills	Priority list items flagged contextually for immediate missing skill study.	Spring Boot, REST API, Hibernate

4. Compliance, Interface Layout & Accessibility Standards (Step 5 & 6)

All interactive web layouts enforce explicit accessibility rules ensuring consistent presentation context without layout shifting or broken text streams:

Design Rule Directive	System Implementation Specification
Restrained Presentation Style	Uses clean structural layouts with an active blue primary accent line. Avoids unnecessary decorative elements to minimize cognitive overhead.
Clear Empty State Logic	When underlying database rows return null values, templates override raw layouts using explicit th:if blocks to prompt clean helper tips.
Meaningful Form Semantics	Mandates strict descriptive markers. Every single input field binds to an explicitly visible HTML label element, ensuring keyboard navigation support.
Multi-Dimensional Explanations	Ensures evaluation outputs communicate score metrics comprehensively via descriptive state fields ('Developing' or 'Expert') alongside raw numerical fractions.